Final source: five reducers and stock CLI

Initialization improves; warmed performance remains slower. At the diagnosed Zggg point TIRust/NinjaRS has matched-cycle median ratio ×**1.553**. The separate stock CLI gives ×1.605. Neither closes local runtime acceptance.

QCD $u\bar{u} \to Zggg$: one diagnosed physical point

Actual `loop_sm` CLI export; final source `77eb57b8`, library `d427`. This follow-up selects **point 1** of the original frozen 100-point set, after its initialization diagnosis. It is separate from the original ten-point timeout. Seven balanced cycles give 35 warmed full-amplitude batches, each one evaluation; actual batches span 3.190–5.712 s.

Table 1: Unscaled MAD of seven times; ratios are medians of matched within-cycle ratios, not ratios of medians. READY separately measures bootstrap plus one same-point warmup. No confidence interval is claimed.

Reducer	Median ± MAD ms / point	Paired ratio / NinjaRS 8	READY s	3e−6 / 1e−8
CutTools	3695.587 ± 57.175	×1.048	7.022	PASS / PASS
Ninja	3575.994 ± 84.260	×1.007	6.627	PASS / MISS
COLLIER	3837.506 ± 127.229	×1.099	7.497	PASS / MISS
NinjaRS	3526.731 ± 82.681	×1.000	6.775	PASS / MISS
TIRust	5426.217 ± 192.886	×1.553	14.119	PASS / MISS

All 45 retained comparisons pass `3e-6`. The separate `1e-8` target misses the finite coefficient for 6/7/8/9 versus CutTools by $\approx 3.534 \times 10^{-8}$ CutTools is its own reference. Old genuine-QP and the other reducers corroborate the cluster, without assigning automatic fault or absolute truth. No point is dropped.

Every reducer has seven accepted DP-transport dispatches. TIRust has **zero genuine Fortran-QP** dispatches but observed local tensor tiers. Timed primary routes are 7448 direct, 392 method-choice, 238 local106 and 42 local256; independent checks are 7518/350/182/70. Attempts106/256 are primary 252/42, check 182/70. Local512, refusal and unreported counts are zero. These counters exclude Fortran coefficient-cache hits. Local tiers lift 53-bit inputs with DP numerators; **219 is not a native-DP-only label**. Getter queries and writes are outside timing.

GNU Fortran 15.2.0; original model/math objects retained, usual generated sources without an optimization flag and polynomial source at O3 (exact recipes retained). Single CPU 12 affinity on a shared, unreserved Linux host; sampled one-minute load 288.1–314.9. This is a selected-point pilot, not a process average, full 100-point campaign or baseline-chain comparison.

Corrected stock `check profile`: QCD $u\bar{u} \to Z$

Table 2: One stock CLI run per backend; unequal short batches, unpaired and descriptive. FULL–SKIP estimates reduction cost; it is not a measured kernel time.

Reducer	FULL ms	SKIP ms	FULL−SKIP estimate ms	Points	FULL wall s
NinjaRS	0.077811	0.065435	0.012376	4117	0.341
TIRust	0.124896	0.064709	0.060187	1007	0.143

All configured helicities, full finite+UV+IR, pole mode 0 and CTModeRun 1. FULL now uses monotonic child-completion timing independent of RSS polling; SKIP/boot retain Python wall clocks. Stock values subtract boot time and divide by actual stage counts. Only the retained all-helicity end result establishes nonzero Born/finite; per-timed-point statuses and tensor routes are unavailable. Missing route counts are **not zero**. The configured polarized result is zero. Separate ten-point within-backend stability probes have 0 QP/unstable/exceptional; maximum spreads are 2.1200×10^{-14} / 1.6033×10^{-15}, not independent error. NinjaRS fresh initialization is 0.05830 s; TIRust reused its export, so initialization is **unavailable (null)**.

Initialization repair and retained precision limits

Final artifact and prototype timings have distinct identities. Final d427 passes 16 native tests, genuine 113⟶256 QP/poles/route smoke and 67 root integration tests. Route-aware tools add 50 tests and a 52-call tensor toy parity check, not another SM/performance qualification. Backend 9 stays explicit.

Quiet initialization: original versus adaptive prototype

The same original O0 Fortran executable, frozen point, cards, stability policy and CPU 12 affinity are used in one ordered old⟶prototype pair. No shim, preload or counter query; other owned captures were held. Shared CPUs were not reserved.

Table 3: Existing Fortran SYSTEM_CLOCK READY: bootstrap plus one same-point warmup, excluding launch and SETPARA. One descriptive pair, not warmed milliseconds/point or a confidence bound.

Library	READY s	Ratio / old	Outer result
Original signed S /85ee	255.241295	×1	323 · genuine Fortran QP
Adaptive prototype /470	15.345797	×0.06012	219 · DP inputs, local tensor tiers

The observed initialization ratio is **16.633**× old/prototype. Final d427 has its own 14.119 s READY above; the prototype timing is not relabeled. The original ten-point attempt's 600 s timeout remains on page 45.

Separate functional observer and tensor reference

The prototype observer overlapped background validation; its elapsed time is excluded. Of 4640 full-DP bridge requests (2320 primary / 2320 check), final routes are 4276 direct, 212 method-choice, 120 local106/DD, 32 local256; no 512/refusal/ unreported or genuine-QP entry. Attempts106/256 are primary 72/12, check 52/20; four 106 attempts continue to 256. Context-lifetime counters survive cache clears and omit pre-validation errors and Fortran coefficient-cache hits.

All comparisons with old true QP and reducers 1/6/7/8 pass `3e-6`; old QP and 6/7/8 also pass `1e-8`. Maximum candidate/old-QP residual is 1.6830×10^{-11}. The CutTools finite target miss 3.534×10^{-8} was already 3.532×10^{-8} against old QP. Genuine Fortran QP retains 113-bit input /256-bit work; local tiers use 53-bit inputs and a DP numerator. Numerical-body identity and final native qualification are recorded separately from prototype observations.

Eight exact-bit tensor fixtures compare against a **same-engine Float512 reference** on 53-bit inputs. All 176 degree×Laurent vectors pass `1e-8` (maximum 9.2988×10^{-17}); actual C-ABI fixture routes are 7 local106 + 1 local256. **50 individual component-relative misses remain** near UV zeros: absolute residue≤8.7664×10^{-27}, reference norm≤2.0021×10^{-146}. All 3136 complex differences are retained; no norm floor. Vector agreement is neither universal componentwise agreement nor an independent physics oracle.

Instrumented attribution; O2 remains unpromoted

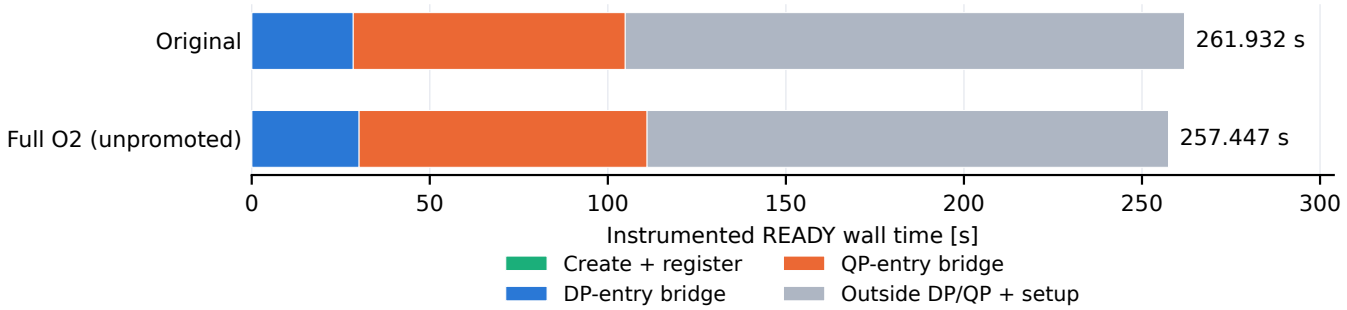


Figure 1: Original/O2 instrumented controls. Remainder is outside DP/QP+setup, not pure Fortran or clean kernel cost; it includes untraced/ boundary work, scheduling and observer overhead.

Original READY 261.932 s contains setup 0.039, DP-entry 28.468, QP-entry 76.383, remainder 157.041 s; first DP⟶QP gap 69.761 s. Its 152 DP refusals lead to whole-QP retries; the separate functional observer accepts 152 local tensor rescues. This attribution is not a phase decomposition of the quiet pair. Full-O2 READY is 257.447 s (×0.983); literal outputs and metadata/status fingerprints match, without full-input proof. No convincing single-control benefit; O2 is unpromoted.

Acceptance remains open. All 66 fresh source-matched numerical lanes and eight main runtime gates await Hua-Sheng's local run. TIRust remains slower in both measured warmed comparisons above. The upper 95% paired candidate/ baseline bound≤1.05 for every main process is not established. Stress failures, strict-target misses, old coarse CLI controls and original timeout remain retained.

Current source 77eb57b82d48 / final DSO d427667a0f2f; prototype 607e7f58842b /470d46df9c8f. Instrumented MG cd014592; old stock cafb35d7 /85ee remains separately archived. Signed TIRust 10ce34ed8ad5; exact hashes, nine CSVs, raw points/checks/routes and pure reproduction accompany this appendix. Original 16 MB source archive is reused by hash. Typography/palette: pyAmpliCol 7266c1600f6e, preserved style.